Research Landscape Brief: Battery Materials & Energy Storage

A cited, primary-source snapshot of research momentum, leading R&D organizations, active subtopics, and must-read work — computed from real OpenAlex records, linked so you can verify every figure.

Sample corpus: 12,000 works · Data: OpenAlex (CC0) · Generated July 21, 2026

1. Research momentum (works per year, 2010+)

Year	Works
2017	823
2018	996
2019	1147
2020	1430
2021	1349
2022	1267
2023	1377
2024	987
2025	307
2026	2

2. Leading R&D organizations (by output in this corpus)

- Chinese Academy of Sciences — 798 works
- Tsinghua University — 342 works
- Nanyang Technological University — 327 works
- University of Science and Technology of China — 301 works
- Argonne National Laboratory — 299 works
- University of Chinese Academy of Sciences — 273 works
- Centre National de la Recherche Scientifique — 245 works
- Central South University — 239 works
- Nankai University — 226 works
- Huazhong University of Science and Technology — 201 works

3. Active subtopics

Materials science (11183), Electrode (9515), Chemistry (9439), Nanotechnology (8398), Chemical engineering (6248), Electrochemistry (6019), Energy storage (5726), Battery (electricity) (4365), Composite material (4286), Metallurgy (3911), Lithium (medication) (3829), Engineering (3570)

4. Most-cited work (verify via DOI)

Materials for electrochemical capacitors (2008) — 16,186 citations

Verify: https://doi.org/10.1038/nmat2297

Commentary: The Materials Project: A materials genome approach to accelerating materials innovation (2013) — 12,839 citations

Verify: https://doi.org/10.1063/1.4812323

The chemistry of two-dimensional layered transition metal dichalcogenide nanosheets (2013) — 9,842 citations

Verify: https://doi.org/10.1038/nchem.1589

The Li-Ion Rechargeable Battery: A Perspective (2013) — 9,794 citations

Verify: https://doi.org/10.1021/ja3091438

Nanostructured materials for advanced energy conversion and storage devices (2005) — 8,780 citations

Verify: https://doi.org/10.1038/nmat1368

30 Years of Lithium■Ion Batteries (2018) — 5,967 citations

Verify: https://doi.org/10.1002/adma.201800561

5. Geographic concentration

CN (38383), US (12141), KR (4087), DE (3508), GB (3213), AU (2969)

Method & limits

Computed from a live OpenAlex sample (CC0). This free sample is a focused subset for speed; a paid brief runs the full topic set and adds your-company-vs-peer positioning, supplier/patent cross-signals, funder analysis, and a corrected multi-year trend. We never assert commercial outcomes — only what the publication record shows.

Want this for your exact chemistry, competitor set, or supply chain? Reply and we'll scope a full cited brief.